

HAVRE, MT, USA

WMO: 727770

Lat: 48.543N

Lon: 109.763W

Elev: 788

StdP: 92.21

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 94012

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-29.8	-26.0	-32.3	0.2	-28.9	-28.7	0.3	-25.0	14.3	3.9	12.8	2.9	2.8	230	0.699

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.3	35.0	17.6	33.0	17.1	31.0	16.6	19.6	30.0	18.5	29.3	17.5	28.1	4.6	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.2	12.7	22.8	15.0	11.7	21.2	13.7	10.8	20.5	59.1	29.8	55.1	29.2	52.1	28.1	28.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.6	11.2	9.9	DB	-34.6	37.9	4.5	1.7	-37.9	39.2	-40.5	40.2	-43.0	41.2	-46.2	42.4	(4)
(5)			WB	-33.4	22.0	3.4	2.2	-35.8	23.5	-37.8	24.8	-39.7	26.0	-42.1	27.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.4	-8.0	-6.4	-0.3	6.5	12.0	16.6	21.3	20.3	14.4	6.8	-0.6	-6.2	(6)	
	DBStd	11.96	9.21	8.90	7.93	5.02	4.34	3.57	3.33	3.63	4.51	5.19	7.35	8.40	(7)	
	HDD10.0	2451	557	460	322	123	28	2	0	13	122	321	503		(8)	
	HDD18.3	4566	816	693	577	354	202	74	11	23	131	357	568	762	(9)	
	CDD10.0	1151	0	1	3	19	89	198	351	318	146	24	3	0	(10)	
	CDD18.3	224	0	0	0	0	4	21	103	83	13	1	0	0	(11)	
	CDH23.3	3947	0	0	1	19	145	411	1639	1347	366	19	1	0	(12)	
	CDH26.7	1772	0	0	0	3	36	134	808	646	143	2	0	0	(13)	
(14) Wind	WSAvg	4.5	4.6	4.4	4.6	4.9	4.7	4.4	4.0	4.0	4.1	4.4	4.7	4.7	(14)	
(15) Precipitation	PrecAvg	314	12	8	15	26	49	60	41	33	30	17	11	12	(15)	
	PrecMax	542	57	23	51	73	156	170	185	118	163	63	30	54	(16)	
	PrecMin	173	0	0	0	0	4	6	0	0	0	0	0	0	(17)	
	PrecStd	96	11	6	12	18	37	37	38	30	28	13	9	11	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.9	13.8	20.3	26.0	30.3	34.0	37.5	37.3	33.6	26.2	18.9	12.0	(19)	
		MCWB	5.7	6.5	9.6	12.3	15.3	17.9	18.7	17.9	16.3	13.3	9.5	5.6	(20)	
	2%	DB	8.5	9.8	16.5	22.3	27.2	30.2	35.5	34.7	30.8	22.0	14.9	9.2	(21)	
		MCWB	3.9	4.2	7.8	11.1	14.5	16.8	17.8	17.1	15.5	11.3	7.5	3.9	(22)	
	5%	DB	6.0	7.0	13.4	19.3	24.5	27.7	33.6	32.7	27.8	18.8	11.8	6.9	(23)	
		MCWB	2.4	2.7	6.1	9.5	13.1	16.0	17.7	16.8	14.5	10.0	5.8	2.5	(24)	
	10%	DB	4.0	4.4	10.3	16.3	21.7	25.6	31.4	30.6	24.6	16.0	9.1	4.5	(25)	
		MCWB	1.3	1.3	4.7	8.0	11.7	15.1	17.2	16.3	13.2	8.6	4.4	1.2	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.1	6.9	10.4	13.4	16.4	20.1	22.2	20.3	18.5	14.1	10.6	6.4	(27)	
		MCDB	11.1	12.9	19.5	23.1	27.2	28.9	32.1	32.4	29.1	24.3	18.5	11.4	(28)	
	2%	WB	4.1	4.6	8.3	11.5	15.0	18.3	20.2	18.7	16.5	11.9	7.9	4.2	(29)	
		MCDB	8.0	9.2	15.2	21.4	25.1	27.7	30.3	30.7	27.2	20.3	14.2	8.5	(30)	
	5%	WB	2.5	2.8	6.6	10.0	13.8	17.0	19.0	17.8	15.1	10.5	6.2	2.7	(31)	
		MCDB	5.6	6.5	13.0	18.6	22.7	25.3	29.5	29.3	25.6	17.8	11.2	6.6	(32)	
	10%	WB	1.4	1.4	4.9	8.4	12.6	15.8	17.9	16.9	13.8	9.0	4.6	1.3	(33)	
		MCDB	4.0	4.4	10.0	15.5	20.1	23.5	28.6	28.1	23.8	15.4	8.9	4.4	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	10.8	11.3	12.4	14.0	14.3	14.3	17.3	17.3	16.1	14.1	11.7	11.0	(35)	
		MCDBR	12.1	13.8	17.5	19.8	19.5	18.6	20.9	21.4	22.2	19.2	15.2	12.9	(36)	
	5% WB	MCWBR	8.5	9.2	10.0	10.3	8.9	7.7	7.4	8.2	9.8	10.2	9.1	8.6	(37)	
		MCWBR	11.6	13.4	16.7	18.7	17.2	16.4	17.9	18.5	20.1	18.0	14.4	12.6	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.264	0.269	0.290	0.334	0.355	0.350	0.355	0.376	0.329	0.297	0.268	0.257	(40)	
	taud		2.452	2.494	2.489	2.371	2.356	2.423	2.414	2.330	2.431	2.517	2.509	2.448	(41)	
	Ebn at Noon		825	908	936	917	906	910	899	862	877	857	816	786	(42)	
	Edh at Noon		62	75	89	111	118	111	111	115	94	73	58	54	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.36	2.33	3.61	4.91	5.88	6.58	7.08	5.82	4.19	2.65	1.54	1.06	(44)	
	RadStd		0.10	0.19	0.22	0.24	0.45	0.37	0.39	0.36	0.32	0.21	0.12	0.09	(45)	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (22 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page